

## IMPACT OF CORE MATERIAL COEFFICIENT OF THERMAL EXPANSION (CTE) ON WARPAGE ASSESSMENT OF LARGER FCBGA PACKAGE WITH STIFFENER RING

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### ABSTRACT

Flip chip ball grid array (FCBGA) is one of the most broadly used package types in different types of electronic packages. In a FCBGA package, the die/ silicon is flipped and attached to substrate with help of copper pillars. The material property coefficient of thermal expansion (CTE) mismatch between die/ silicon and substrate materials like core, dielectric material results in package thermal warpage and may lead to thermo-mechanical related failures in electronic packages. CTE can be defined as the rate at which a material expands with change in the temperature. Material expansion increases with a higher CTE. The CTE of die/ silicon is lower than the overall CTE of organic substrate. During reflow, as an electronic package cools down it warps in a convex or crying shape which indicates positive package warpage, and as the electronic package heats up it warps in a concave or smiling shape which indicates negative package warpage. Induced thermal warpage during the electronic package reflow assembly process is an important factor affecting the package assembly and thermo-mechanical reliability of FCBGA packages; therefore, precise package thermal warpage prediction is one of the most significant parameters for the electronic package design processes.

In this study, the effect of different electronic package parameters (core material CTE, core thickness, package type) of the FCBGA package on its warpage are assessed by using finite element analysis (FEA) mechanical simulation methodology. A bare die FCBGA package having substrate core material with lower coefficient of thermal expansion (CTE) shows lower package warpage as the mismatch between die/ silicon and substrate core material CTE becomes dominant factor to drive the package warpage. To understand the package warpage trend, warpage is evaluated at room temperature & high/ reflow temperature. The package warpage trend is reversed for larger FCBGA packages with stiffener ring. Substrate core material with higher CTE shows lower package warpage as CTE mismatch between substrate core material and stiffener ring

material (steel/copper) CTE becomes the dominant factor to drive the package warpage. The counter intuitive package warpage data from mechanical simulation is compared with shadow-moire test data and it shows good correlation and validates the simulations findings. Based on this work, it is recommended that a thorough investigation for large size packages be conducted before choosing packaging materials. This work shall provide new insights for package designers to optimize thermal warpage and stress during package initial design phase.

Key words: Flip chip BGA packages, finite element analysis, thermo-mechanical warpage, shadow-moire deformation, substrate core material, stiffener ring packages, CTE mismatch.

### 1. INTRODUCTION

In today's rapidly evolving technological landscape, electronic devices are becoming increasingly sophisticated and powerful. A crucial component in these devices is the packaging that encases semiconductor chips. Flip Chip Ball Grid Array (FCBGA) packages are widely used due to their efficiency and reliability. These packages involve flipping the chip and attaching it to a substrate, which can lead to challenges arising from differences in how materials expand and contract with temperature changes. Advanced packages consist of multiple components with significantly different thermal, mechanical, and electrical properties, all of which undergo complex manufacturing processes. These processes induce thermo-mechanical stress, posing various mechanical challenges. When materials with different rates of thermal expansion are assembled or processed to form a package, temperature changes can cause thermo-mechanical stresses and deformation. Such induced stress can lead to mechanical failures, including delamination, instantaneous fracture, and fatigue failure. Additionally, it can result in higher warpage or issues with package coplanarity, presenting assembly challenges that may lead to yield loss or reduced package reliability. Figure 1 illustrates the induced

thermal warpage during the package reflow assembly process for conventional FCBGA packages. This factor is crucial in affecting the assembly and thermo-mechanical reliability of FCBGA packages. Recent advancements in Artificial Intelligence are driving the demand for accelerated computing and higher bandwidth memory, necessitating more complex and larger packages [1, 2]. These advancements are likely to present additional challenges in terms of yield, warpage, and reliability. This study focuses on the impact of substrate core CTE on warpage for advanced larger packages with stiffener rings, aiming to provide insights into optimizing package design to minimize thermal warpage and enhance reliability.

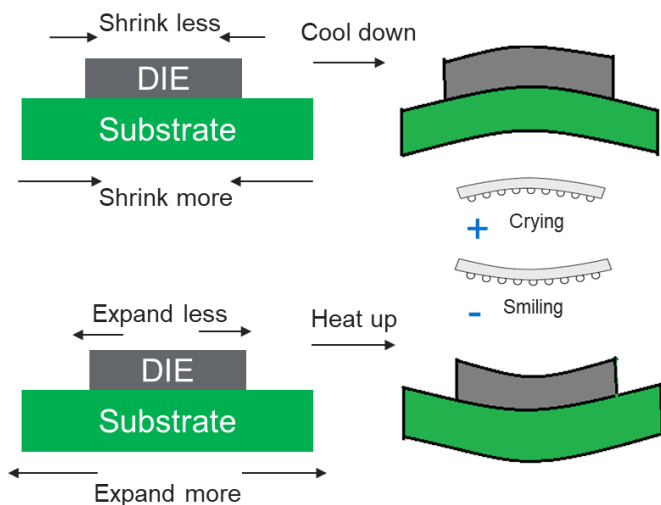


FIGURE 1: SCHEMATIC OF INDUCED THERMAL WARPAGE DURING PACKAGE REFLOWASSEMBLY PROCESS FOR CONVENTIONAL FCBGA PACKAGES (NOT TO SCALE)

## 2. MATERIALS AND METHODS

### 2.1 Materials

For package semiconductor packaging material properties used in this study for different package components are linear elastic material properties. Only for solder ball, viscoplastic SAC305 material properties are used [3] [4]. Substrate is considered layer by layer. Each signal and dielectric layer modeled with volumetric average material properties considering metal percentage in each signal layer [5]. Interconnect and underfill between the die and the substrate are modeled with homogenous block considering the volumetric average.

### 2.2 Modeling methodology

The modeling is done for the stand-alone package before attaching it to printed circuit board. The simulation considers major assembly process impact such as die attach and stiffener ring attachment. The major mechanical challenges considered are the package coplanarity and thermo-mechanically induced stress.

## 2.3 Package configurations

A stiffener ring package structure with a substrate size exceeding 70x70mm, incorporating a System on Chip (SOC) die equivalent to the reticle size, was considered for this study. The package is constructed using three different core materials: Low, medium & high CTE. Package total thickness is approximately 5mm. FIGURE 2 shows schematic of typical stiffener ring package.

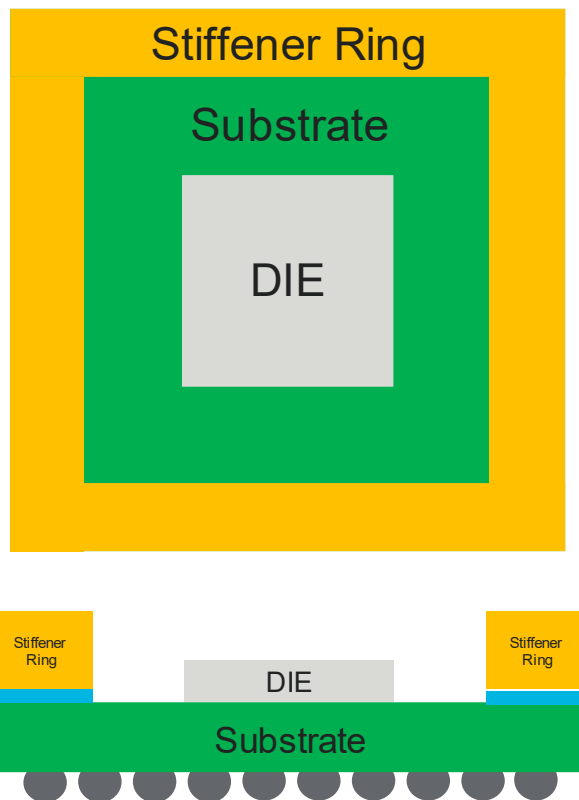


FIGURE 2: SCHEMATIC OF STIFFENER RING TYPE PACKAGE, TOP VIEW AND SIDE VIEW (NOT TO SCALE)

## 3. RESULTS AND DISCUSSION

### 3.1 Model correlation and benchmarking

The Ansys model is benchmarked with empirically measured package deformation with shadow moire. The out of plane deformation at the bottom of the package is measured (also commonly referred as package warpage) using shadow moire. The model is predicting deformation with 10% accuracy over the temperature range. The shape of the deformation at room temperature (RT); ~25C is convex and shape of the warpage deformation at the high temperature (HT); ~260C is concave. FIGURE 3 shows Ansys warpage model benchmarking with shadow-moire data. Same modeling methodology is used to model 70x70mm package with stiffener ring structure with three different core materials. The model predicts similar deformation shapes, showing good correlation with empirically measured package warpage with 10% accuracy.

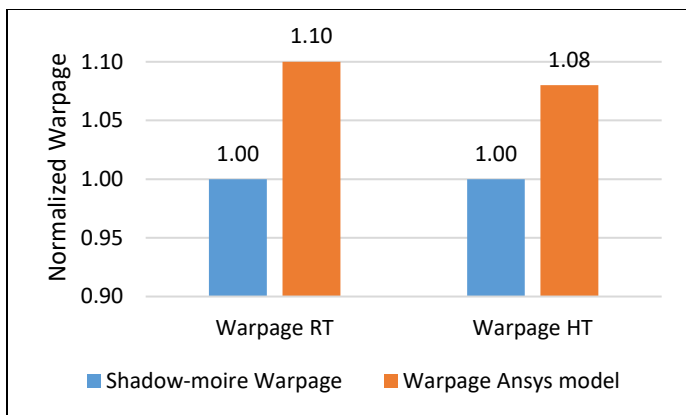


FIGURE 3: ANSYS WARPAGE MODEL BENCHMARKING WITH SHADOW-MOIRE WARPAGE DATA

### 3.2 Substrate core materials

FIGURE 4 shows the thermo-mechanical properties (CTE) for three different substrate core materials. Substrate core material 1 is used as the base case, and the CTE for other materials is normalized with respect to it to highlight the changes. FIGURE 5 shows the impact of the substrate core material CTE on package warpage. Substrate core material 2 has an 11% lower CTE compared to material 1, and core material 3 has a 32% lower CTE compared to material 1. The packages are built using all three core materials, with the only difference between them being the core material used, while all other package parameters remain the same. Shadow-moire warpage data is collected for all three package configurations to understand the impact of substrate core CTE on large packages with stiffener ring structures. Ansys warpage simulations are also completed to understand the correlation between the Ansys model and experimental shadow-moire data. The Ansys model correlates very well with the shadow-moire data, with 10% accuracy.

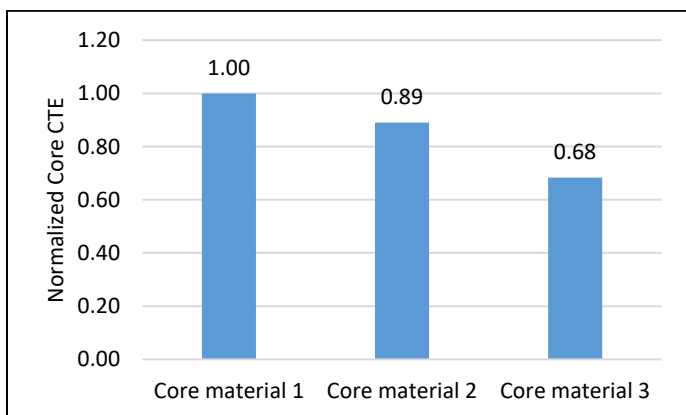


FIGURE 4: THREE DIFFERENT SUBSTRATE CORE MATERIAL MECHANICAL PROPERTIES (CTE)

### 3.3 Impact of substrate core material on package warpage: Measured shadow-moire test data

FIGURE 5 shows the impact of the substrate core material CTE on package warpage at room temperature (RT) and high temperature (HT). Substrate core material 1 is used as the base case, and shadow-moire warpage is normalized for core materials with respect to it to understand the impact on package warpage when the CTE of the substrate core material changes from high CTE to low CTE. FIGURE 5 illustrates the change in shadow-moire warpage at RT and HT with three different core materials. When the core CTE is reduced by 11%, there is a 1% increase in warpage at RT and a 14% increase at HT. Similarly, a 32% reduction in core CTE results in a 17% increase in warpage at RT and a 64% increase at HT.

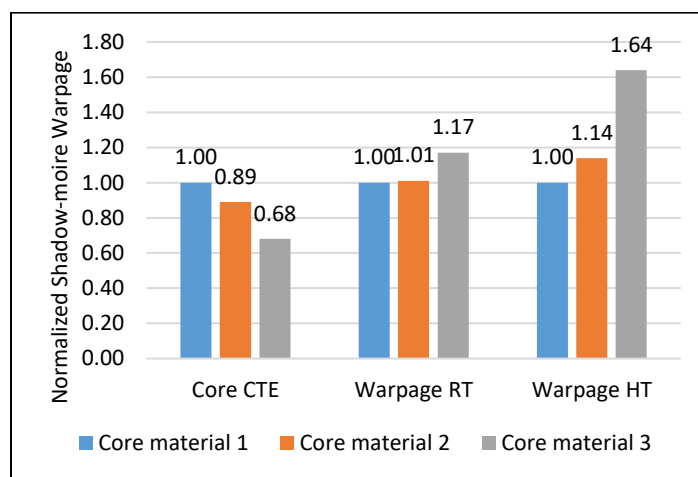


FIGURE 5: SHADOW-MOIRE TEST DATA: IMPACT OF CORE MATERIAL CTE ON LARGE PACKAGE WARPAGE WITH STIFFENER RING

### 3.4 Impact of substrate core material on package warpage: Ansys model data

Ansys warpage simulations were conducted for the same three package configurations to determine if the model exhibits a similar trend to the measured shadow-moire test data. This analysis aims to understand the correlation between the Ansys model and shadow-moire data, ensuring that the Ansys model accurately predicts warpage trends and can be utilized for future similar package configurations. FIGURE 6 shows the warpage simulation results from the Ansys model. Substrate core material 1 is used as the base case, and Ansys model warpage is normalized for core materials with respect to that and compared to understand the impact on package warpage when the CTE of the substrate core material changes from high CTE core material to low CTE core material. Figure 6 illustrates the change in Ansys model warpage at RT and HT with three different core materials. When the core CTE is reduced by 11%, there is a 23% increase in warpage at RT and a 12% increase at HT. Similarly, a 32% reduction in core CTE results in a 41% increase in warpage at RT and a 28% increase at HT. The warpage trends at

RT and HT from the Ansys model closely match the measured shadow-moire test data. However, there is some discrepancy in the magnitude of warpage change at RT and HT between the Ansys model and the shadow-moire test data. Further refinement of the Ansys model is required to reduce this discrepancy.

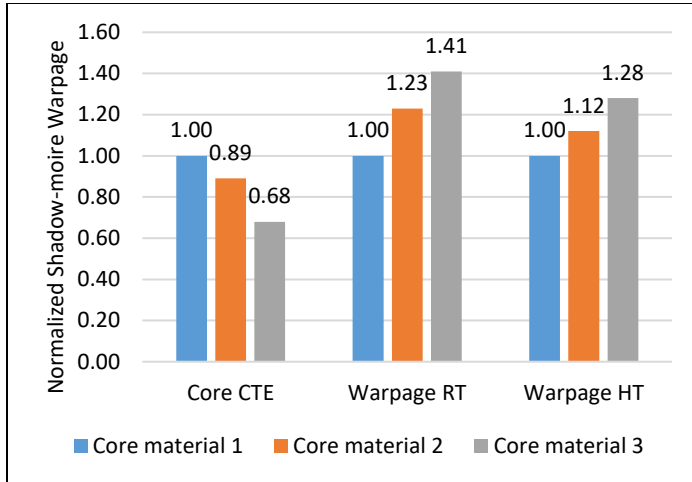


FIGURE 6: ANSYS MODEL DATA: IMPACT OF CORE MATERIAL CTE ON LARGE PACKAGE WARPAGE WITH STIFFENER RING

#### 4. CONCLUSION

This study investigates the impact of three different substrate core materials, ranging from low to high CTE, on the warpage of large packages with stiffener rings. The findings indicate that the substrate core material's CTE significantly affects package warpage. Packages with higher CTE cores exhibit lower warpage, while those with lower CTE cores show higher warpage at both room temperature (RT) and high temperature (HT). This trend is contrary to the conventional package warpage trend. For conventional FCBGA bare die packages, the mismatch between the die and substrate core CTE is the dominant factor driving warpage. However, in larger packages with stiffener ring configurations, the mismatch between the substrate core and stiffener ring material CTE becomes the dominant factor due to the stiffener ring's larger surface area on the substrate compared to the die. Substrate cores with lower CTE exhibit higher warpage due to the larger mismatch with the ring CTE compared to substrate cores with higher CTE. This study demonstrates that a lower CTE core does not necessarily result in lower warpage for larger packages with stiffener rings. In summary, the study emphasizes the need for rigorous design investigation of package structure and material selection to improve the yield and reliability of advanced, larger, complex packages with stiffener rings.

#### REFERENCES

- [1] M. Technology, "Marvell Blogs," Marvell Technology, 14 May 2024. [Online]. Available: <https://www.marvell.com/blogs/the-ai-opportunity-at-marvell.html>. [Accessed 06 May 2025].
- [2] Nvidia, "GTC March 2025 Keynote with NVIDIA CEO Jensen Huang," Nvidia, 18 March 2025. [Online]. Available: [https://www.youtube.com/watch?v=\\_waPvOwL9Z8](https://www.youtube.com/watch?v=_waPvOwL9Z8). [Accessed 06 May 2025].
- [3] S. Ramdas, P. Rajmane, T. Chauhan, A. Misrak and D. Agonafer, "Impact of Immersion Cooling on Thermo-Mechanical Properties of PCB's and Reliability of Electronic Packages," in *ASME 2019 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems*, Anaheim, California, 2019.
- [4] S. Ramdas, A. Chowdhury, A. Lakshminarayana, R. Bhandari, A. Misrak and D. Agonafer, "Impact of thermal aging on thermomechanical properties of oil-immersed printed circuit boards," in *SMTA International Conference*, Rosemont, Illinois, 2019.
- [5] S. S. Ramdas, T. Chauhan, M. Patterson, D. R. Shirley, L. Zhang and R. Coccioli, "Minimizing Die Cracking Risk by Copper Pillar Design Optimization for Advanced 5nm Silicon Node," in *International Electronic Packaging Technical Conference and Exhibition*, San Jose, 2024.